

[Techstack] Rules

The tech sector is a vast capitalist wilderness waiting to be occupied by the best and brightest in the business. Can you corner the market on emerging technologies and profit as your rivals stack solutions on top of them? Or will their teams of expert developers leapfrog your basic building blocks to release more compelling and lucrative commercial projects while yours founder in development?

Move fast to build valuable technology, but be wary of giving away too much to venture capitalists or neglecting team morale and brand identity!

The game is played over 7 rounds, with checkpoints after rounds 3 and 5. The goal is to acquire  victory points by developing and improving technology stacks, releasing projects which use stacks owned by you or your rivals, growing a talented team, and securing track bonuses, card effects, and positive cash flow.

Each round includes two larger phases: In the **Action** phase, players recruit developers, play industry cards, use special actions and initiate the development of projects on their player board. In the **Release** phase, developers code improvements in projects and those which complete can be released into the game – *if* you can meet their technology and Go-To-Market requirements.

Key Concepts

Stack technologies

-  **Operating System**
-  **Database**
-  **Web Services**
-  **Digital Media**
-  **Machine Learning**

Category tags

-  productivity
-  e-commerce
-  social media
-  entertainment
-  artificial intelligence

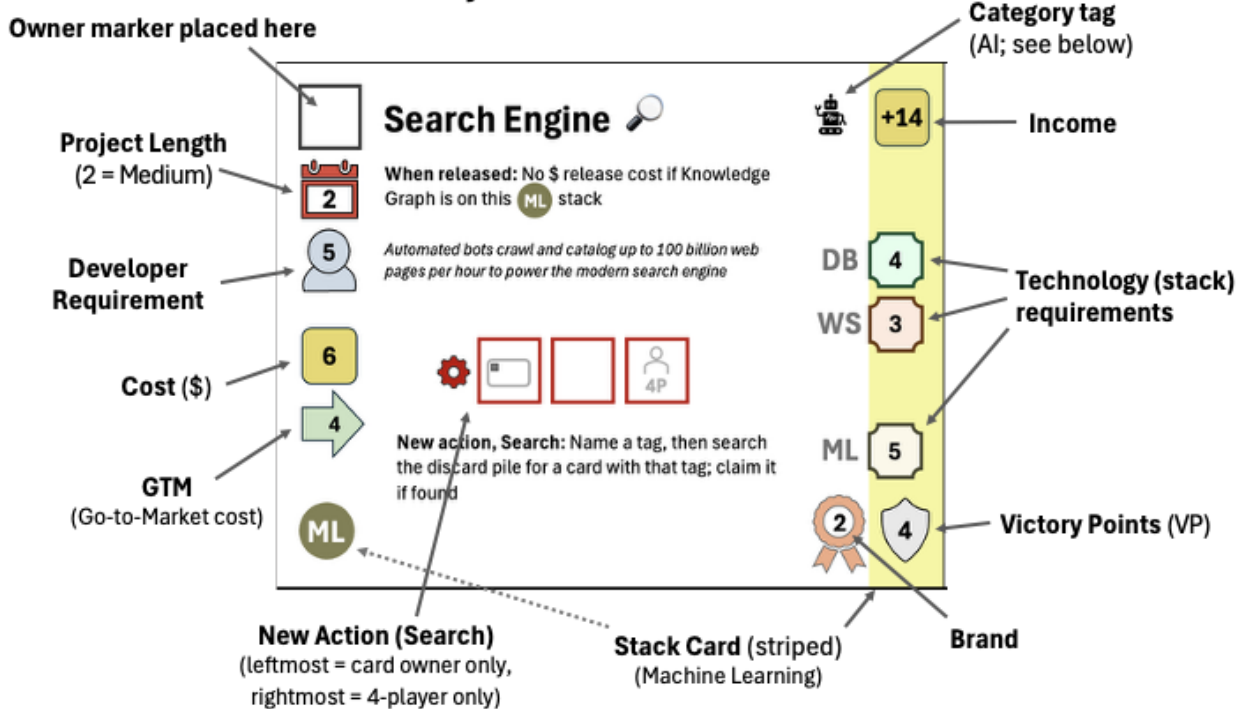
Values / resources

- **Money**  and **Income** 
- **VP**  – victory points
- **Dev strength**  – represents both requirements on project cards and the strength a developer card provides to meet those requirements and build projects
- **GTM**  – Go-To-Market values represent units of company effort (in marketing, engagement, customer support and documentation) required to release a project
- **Brand**  – the brand value of your company; higher brand increases GTM, extends card choice and grants bonus VP
- **Morale**  – the engagement and energy level of your company's staff
- **Technology requirement**  – a required level of a specified stack technology

Terminology

- **Technology** – the five types of stack technologies
- **Project card** – any software card that you must develop before releasing/playing (horizontal / landscape orientation)
- **Industry card** – all other cards in the tech deck, which are played directly from your hand (vertical / portrait orientation)
- **Stack** – a stack of project cards associated with one of the 5 technologies; the stack itself is owned by the player who started it but can include projects owned by rivals. Each stack can be used to satisfy requirements for its technology type as new projects are released.
- **Stack area** – central area of play for card stacks
- **Stack card** – a striped project card that is played into a stack (adding to the stack's level or abilities)
- **Base stack card** – a card which begins a stack, either a base project (such as “Operating System”, starting at level 1) or a New Stack card
- **New Stack card** – a base stack card that builds on the level of on an existing stack
- **Technology level** – the current capability level of a stack (1-10+)

Project Card Reference



Components

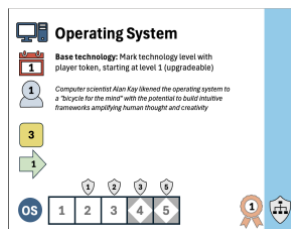
- 6 boards: market board, round tracker board, 4x development boards (1 per player)
- Player tokens, per player: 20 color markers (cubes), 12x each small and senior (large) developer tokens, 1 recruiter token, 1 income tracker, 1 staff cost tracker, 1 morale marker, 1 brand marker
- Money counters, GTM (Go-To-Market arrow) counters, starting player token, round marker
- Venture Capital (VC) funding cards – 3 per player (Series A, B and C – 12 total)
- Developer deck – representing advanced developers that players hire for special abilities
- Tech deck
 - Project cards – software projects players can build on their development board and then release into active play for income, effects and/or VP. **All cards in landscape orientation are project cards** which must be built before they can be played.
 - Industry cards – a mix of special staff, event and investment cards with costs and benefits that can be paid for and played immediately. Industry cards are in profile (upright) orientation.
- Base projects (fixed set) – 2 copies each of 5 upgradeable project cards (1 for each stack technology)
- Standard projects (3x per player) – useful projects that are permanently available to players
- Checkpoint cards (10) – analyst review bonus cards with cost-of-success setbacks on the reverse (full game)
- Community Upgrade cards (10)



Developer tokens



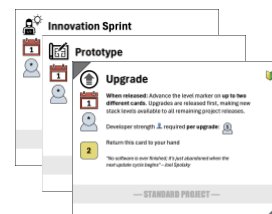
Go-To-Market (GTM) counter



Base project (OS, 1 of 10)



VC funding cards (one set per player)



Standard projects (one set per player)

Setup

The basic game 📖 plays more quickly and with fewer challenging obstacles – recommended for your first game.

Place the **round tracker** and **market boards** at the top of the play area. For the **basic game**, use the boards marked with 📖.

- **Morale 📖 and brand 🏆 tracks** – place player markers on the starting spaces, marked with a ☆
- **Developer talent row** – shuffle the developer deck and lay out a row of 6 cards face up in the top row of the market board (marked with 👤) to form the **talent row**. Place the deck face down on the right.
- **Card market** – shuffle the main (tech) deck and lay out a row of 6 cards face up along the bottom of the market board to form the **card market**; place the remaining deck face down to the right. Both card rows always immediately refill from the right, off the top of their respective decks. Leave room for a discard pile next to the deck.
- **Round tracker board** – place the round marker on round 1, place each player's recruiter token on the blue action space marked with a ☆, and place player trackers starting at 6 and 0 for staff costs and income, respectively.
- **Full game only:** randomly draw and place checkpoint cards next to the indicated spaces after rounds 3 and 5 on the round tracker ('analyst review' side up, showing a tag bonus and a ❤️ warning.)
- **1 or 2-player game only:** Add starting neutral stacks by randomly selecting base project cards and placing them in a central stack area with a neutral player marker at level 1. In the solo game, select 2 (different) base projects in this way and in the 2-player game select 1.

Per player

- Each player chooses a color along with the associated player markers and tokens
- One **development board** (player board), each with four available spaces with sliding positions for projects in development
- **3 VC funding cards** – each player receives a Series A, B and C card, tucking them under the top of their development board; these cards may be flipped (in order, starting with series A) to accept Venture Capital funding (at the cost of victory points shown on the front of the card).
- **\$20** in starting money (**Full game: \$10**)
- **1 GTM counter** (arrow), matching the single GTM ➡ value printed on each player board
- **2 small developer tokens** (starting team)
- **3 standard projects** – Upgrade, Prototype, Innovation Sprint. For the **basic game**, use the Upgrade cards marked with 📖

Starting cards

- Players keep two hands of cards: a player hand (for tech cards), and a developer hand, a set of developer cards you keep to the side and use to gain special abilities as your team becomes more skilled.
- For the **basic game** 📖: Separate and randomly deal **1 starter project** (from the tech deck) and **2 starter developer cards** to each player. These cards are marked by triangles in the upper left and lower right corners:



Shuffle any remaining starter cards back into their respective decks.

- **In the full game**, deal 4 developer cards to each player; players keep 2 and discard 2
- Starting hand – deal each player up to 8 tech cards (counting any starter project 📖); players keep 4 in total and discard the remainder

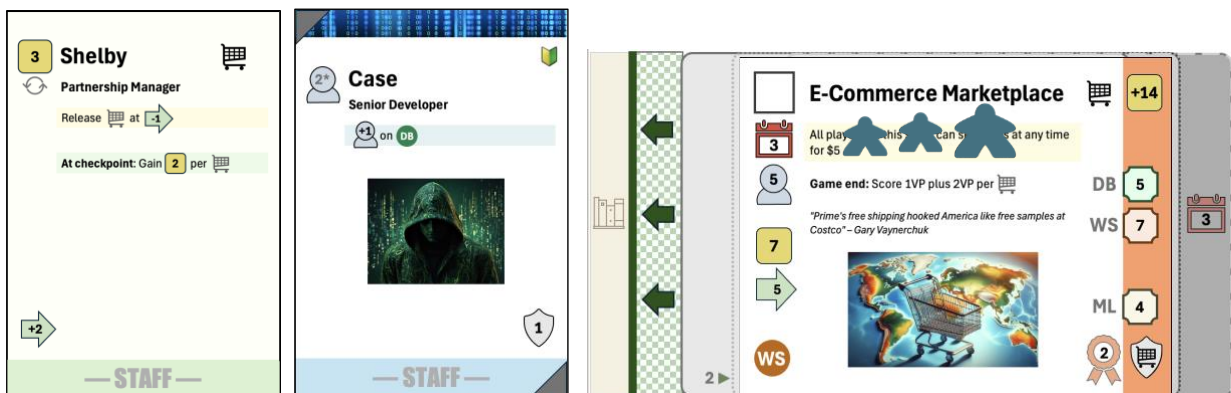
Give the **starting player token** to the player who least recently looked at their phone. Play proceeds in clockwise order (to the player's left)

Gameplay

The game is played over 7 rounds, with checkpoints after rounds 3 and 5. Each round, players gain income, pay staff, and claim new tech cards. Then, players take turns during two major game phases: Actions, during which players recruit, play industry cards and make use of additional actions as they are introduced into the game by stack cards; and Release, during which players release new projects they've developed and resolve associated effects and bonuses.

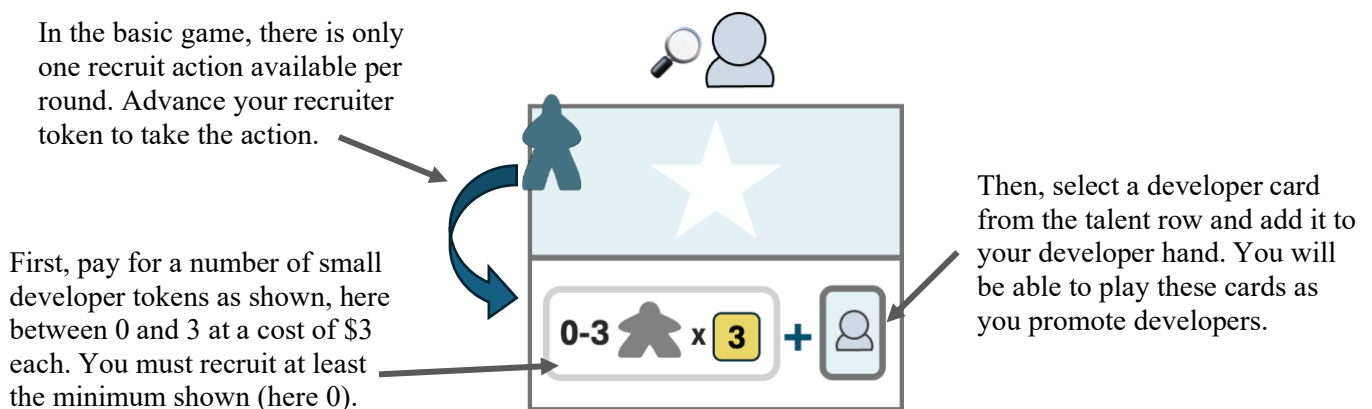
Round structure: (*first round only: skip ahead to Planning*)

1. **Income** – all players update their income tracker: add **card income** (all  values as printed on cards), to **stack income** (the count of your markers stored by all players on their development boards). Gain this income.
2. **Staffing** – all players:
 - a) **Refresh** – if the leftmost card in the talent (developer) row has a coin on it, discard the card and coin and refill. Place a coin on the leftmost developer card.
 - b) **Retention** – for each  symbol indicated by a player's morale tracker, randomly select one staff member to release (lose) as attrition. First make note which senior developer tokens go with which played developer cards if necessary, then ask your phone for a random number, counting each developer token and industry staff card as eligible. Return any released developer token to your player pool, but do not yet discard any released staff card – see 'Recruit free agents' below. (Do not pay costs for lost staff members this round or lose additional .)
 - c) **Downsizing** – players may choose to **release staff members** of their choice to reduce costs – either developers represented by a token plus any associated senior card, or industry staff cards. -1  per staff member released in this way. Return any released developer tokens to your player pool, but do not yet discard any released staff card – see 'Recruit free agents' below.
 - d) **Bid for free agents** – For each staff card released in (b) or (c): Any other player may claim the released staff card if they are willing to pay its cost this round. If more than one player is interested, they may bid (starting at or above the card's printed cost), in player order until one bidder remains. The winner claims the card and plays it into their play area, immediately resolving any “when played” or “at promotion” card effect. If the card is a senior developer, the winner takes an associated senior developer token. Any “once per game” developer abilities are renewed. If no players are interested in a released staff card, discard the card.
 - e) **Payroll** – all players update their staff cost tracker, showing total recurring staff costs: add developer costs (**\$3 per small developer token plus \$5 per senior developer token**) to staff card costs (all  values as printed on industry staff cards). Pay this total cost.

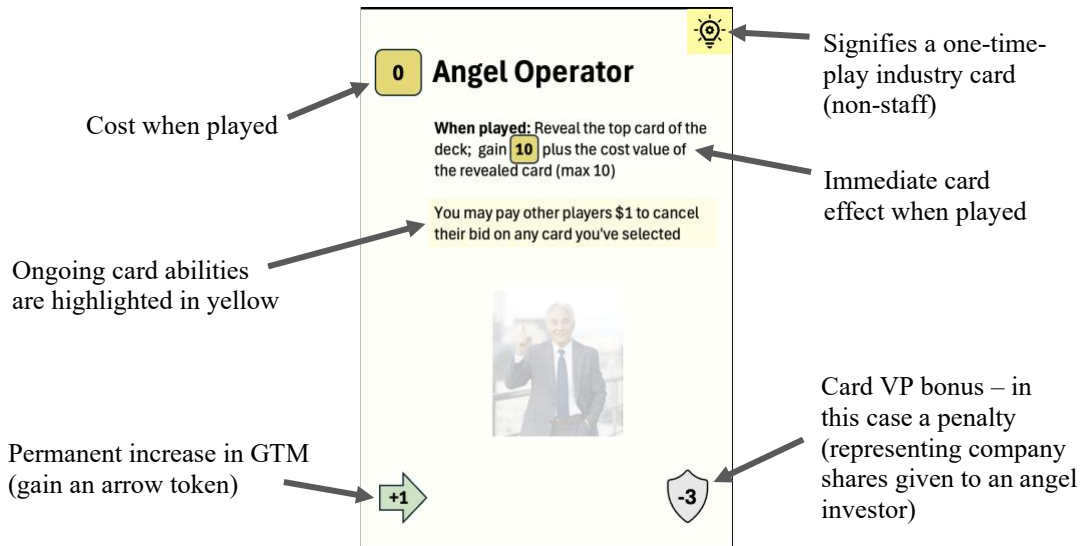


In this example, the blue player has \$14 in staff costs showing: \$6 for two junior developers, \$5 for senior developer Case, and \$3 in industry staff costs for Shelby

3. **Open source** (*1-3P only*) With each open source update, the lowest-leveled technology will increase. In a 3P game, generate one update; in a 1 or 2 player game generate two. For each update: if there is a type of technology missing, add a base project card (in priority order OS, DB, WS, DM, ML) and place a neutral player marker at level 1. Otherwise, increase whichever technology is at the lowest level, either by advancing the level marker on a neutral-owned base project, or by sliding a **Community Upgrade** card under a base stack card owned by an active player. If there is a tie between technologies (that do not have any stack above a certain level), use the same priority order listed above. If a technology has multiple stacks to choose from, increase the *highest* leveled stack (the community wants to increase capability). If there is a tie between stacks within a technology, resolve in favor of neutral stacks over player stacks, and then in favor of players who have the fewest total stack markers on player boards.
4. **Planning**
 - a) **Refresh** – if the leftmost card in the market row has a coin on it, discard the card and coin and refill the row. Place a coin on the leftmost card. (*Solo game only: instead, discard one card from the spaces in the market that are available to you.*)
 - b) **Card market** – in order, players **select a card from the market row** and refill. You may select from as far into the market row as your brand tracker reaches (shown above the market row; at the start of the game this means choosing between the two leftmost cards.)
 - i. After a player selects a card, others may indicate they wish to **bid** for it. Starting on the player's left each player must bid or pass (and sit out the remainder of bidding) until a winning bid is determined. If there are no bids the player takes the card without cost. If another player wins the card, they pay the bank and the original player selects again.
 - ii. **Each player will draft one card** in this way and sit out any remaining bidding. Refill the market row after each card is selected (slide cards to the left and refill from the top of the deck).
 - iii. As an alternative to claiming a card in the market, players may choose to discard an available card from the market and instead draw the top card of the deck into their hand (blind draw).
 - c) **Assignments** – assignments of developer tokens to projects on player development boards may now be rearranged to start the round. Any changes in assignments must continue to meet project card requirements, or the project cards must be returned to the player's hand (players may also choose to do this voluntarily, returning a project card to their hand and freeing up the development space and assignments).
5. **Actions** – in turn order, each player takes **one or two** of the following actions, until all players have passed.
 - a) **Recruit** – advance your recruiter token one position on the round tracking board, and gain the options shown for adding developer talent to your team. First, pay for new junior (small) developer tokens at the cost shown (e.g. if the cost is \$3 and you take two new tokens, pay \$6 to the bank), and place these tokens in your play area. Then, select a developer card from the talent row, add it to your developer hand, and refill the row. You may select from as far into the talent row as **either** your morale or brand tracker reaches.



- b) **Begin development** – Select one or more project card(s) from among your standard projects, cards in hand **or** available base project cards, then place them in any available development spaces according to each card's project length 📅 (1-3 rounds). Assign developers by placing developer tokens on each project card: small tokens have a dev strength 🧑 of 1, senior tokens are worth 2. The combined strength of tokens must meet or exceed each project's 🧑 requirement. When placing a senior developer token, place the associated senior developer card next to the assigned project. *Full game: developers remain on their assigned project for the remainder of the current round and may not be reassigned.* **Important restriction:** players may only begin development on **one** base project (e.g. "Operating System") per game **turn** (that is, including both actions.)
- c) **Play industry card** – Pay the card cost and play face up in your play area. Resolve "when played" card effects. Note that "Staff" card costs recur and will need to be paid during Payroll in future rounds.

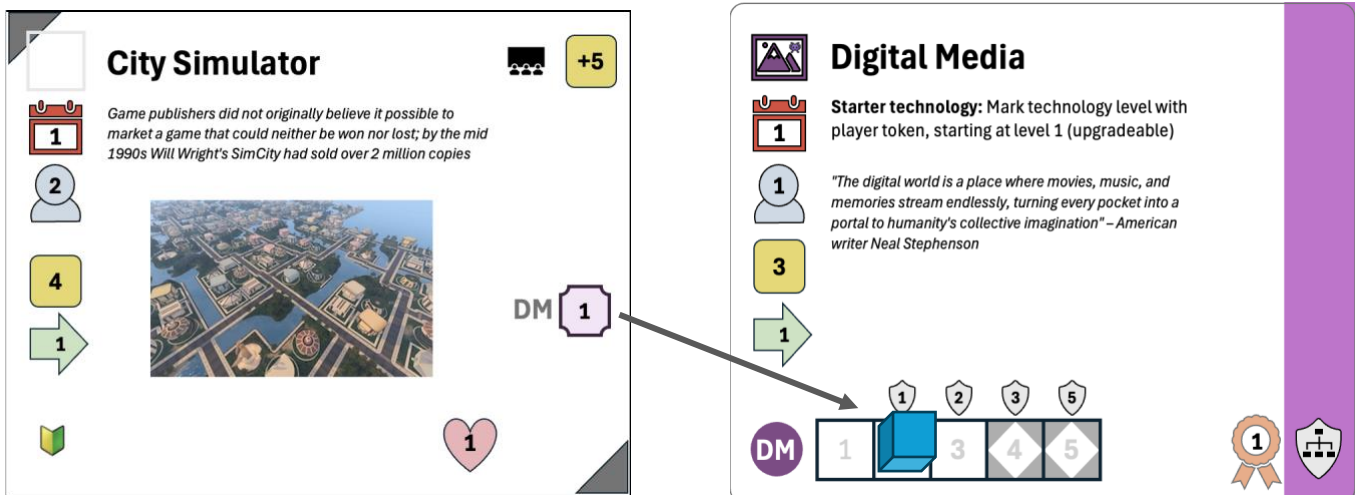


- a) **Use card actions** ⚙️ – as described in a card's "new action:" text. In the case of stack card actions, you must be "on" the stack in question (have a player marker for that stack on your development board, see Release below).
- b) **Right-size** – discard 2 cards for +1 📄
- c) **Free actions** – as described on played cards (free actions do not count as a separate action and may be taken before passing.) Note players may also accept or pay off VC funding cards at any time (without requiring an action, and during any game phase).
- d) **Pass** – first player to pass may swap any 2 cards in the card market (2-4P only, move coins with cards.)

6. Projects

- a) **Advance** – Time passes, and developers code! Check that all projects have appropriate 🧑 dev strength assigned and slide them one position closer to release. Projects which reach the release line become **release candidates** (actual release remains subject to technology requirements and paying \$ and GTM ➡️ release costs).
- b) **Promote** – Each player promotes **one** eligible developer assigned to a project
- Players may choose to skip an eligible promotion (avoiding the increased salary cost) at a penalty of -1 📄
 - Replace one small developer token with a senior token. **Assign a corresponding card from your developer hand** to the same project (or to your play area, if the developer token is currently unassigned.) Immediately resolve any 'at promotion' card effect. **Card abilities become immediately available** – for example, enabling a discount on a developer's project this round. Note that the increase in dev strength 🧑 (from 1 to 2) only adds immediate value when a project has variable effects based on dev strength, as with Upgrade and Prototype.
 - Promote in turn order if order matters (e.g. abilities dealing with the discard pile.)

- c) **Release**— in turn, players may release eligible (completed) projects, paying costs and expending GTM capacity as they do.
- First, release all **Upgrade** standard projects. Advance level markers and pay costs per instructions on the card – increases in the levels of stacks become immediately available to satisfy technology requirements on other projects releasing this round.
 - Release all other projects** – in player order, select **one or two projects** to release, verifying requirements and costs:
 - Meet technology requirements** – for each tech requirement on the right side of the card, a stack which meets or exceeds the indicated level must be used (selected). The player **collects a stack (player) marker for each requirement** and adds it to the designated pool at the top of their development board.
- If more than one stack meets a required level the player may choose; however, **they must always use their own stack** if they own a stack of that type
 - Pay **\$** and GTM  costs. Pay money costs to the bank and place required GTM  tokens on the project card to keep track of the Go-To-Market resources used. Players may gift unused GTM counters to other players during Release.



In this example City Simulator has reached the release line; the owner releases it by paying \$4, matching the required 1 GTM  by marking it with an unspent  token, and by using the blue player's Digital Media, which has reached the required level 2. The player adds a blue marker to the **DM** area at the top of their development board and plays City Simulator into their play area. They then gain card rewards: +1 to morale from the card's  icon (gaining team morale, advancing their morale tracker), and advancing their income tracker by 5. The player has also gained an entertainment  tag which could trigger game effects or add to bonuses later in the game.

ii. **Place the played project into the appropriate game area:**

1. **Stack (striped) project cards**

- Base project cards begin a new stack in the stack area.
- New Stack cards build on the level of an existing stack; the releasing player selects a stack that meets this card's requirement for the indicated technology type. If the releasing player also owns the selected stack, the card is placed at the bottom of that stack (increasing its level); otherwise, this card begins a new stack in the stack area. Place a marker for the original stack in the lower left of the card, then mark the first box on the level track with a marker of the new (possibly different) owner. As indicated, the level of the stack is equal to the level of the original stack plus the marked value on this card's level track.
- " +1 " or " +2 " improvement cards slide under base cards of the same type, and add to the level of that stack (see Relational Database in the example below.)
- All other stack cards are placed at the bottom of the stack they are using; these cards add new abilities to the stack (in addition to benefits that may be specific to the card owner). Place the owner's marker in the upper left.

2. **All other cards** are played in each player's play area (unless otherwise noted on the card). Placing a marker is optional; these cards may be stacked to preserve space (but their contents are public information and can be viewed at any time).

- iii. **Resolve "When released" card effects** as each project is released (see clarifications in Glossary, e.g. for Upgrade)

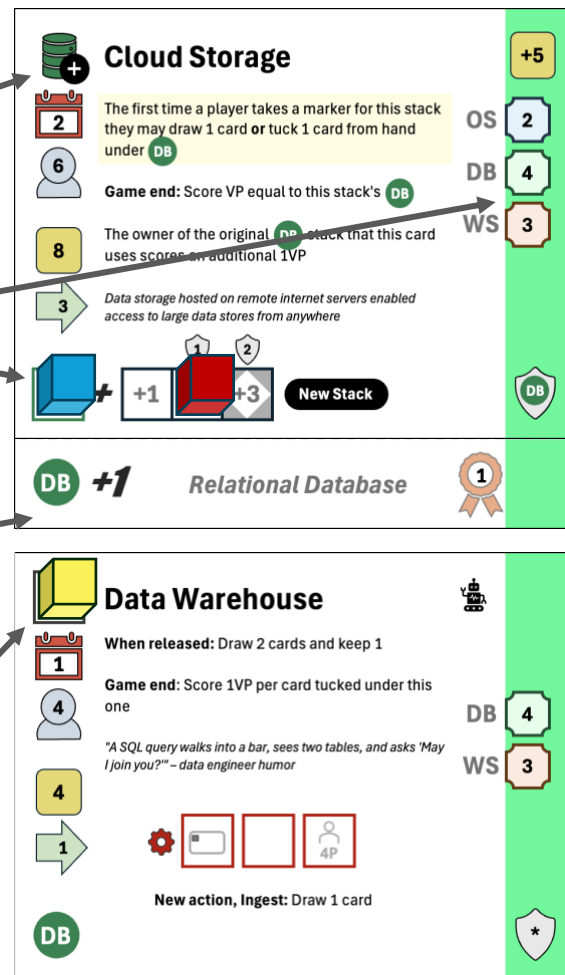
Stack example

Cloud Storage is a New Stack card, so it starts a new stack – a column of stack cards with an associated technology value that can be used to satisfy **DB** requirements

To play Cloud Storage, the red player built on top of the blue player's Machine Learning stack, which was at level 4 and so satisfied the card's **DB** requirement shown at right. Red placed their own player marker on the card's level track.

Relational Database is a stack improvement that has been played on Cloud Storage. This stack's **DB** is now at 4 (from blue's) plus 2 (an upgrade on Cloud Storage from +1 to +2) plus 1 from Relational Database, for a total of 7.

Data Warehouse is a stack card, so yellow has played it in the stack **DB** under Cloud Storage, which easily satisfied its requirement of 4. Data Warehouse doesn't add to the level of this stack, but it provides a new action that becomes available to any player on this stack (a player who has used red to satisfy a **DB** requirement.)



- c. **Continue releasing projects in player order:** players may pass (wait) if their remaining projects rely on or will take advantage of a stack card that another player is eligible to release but has not yet released. The waiting player must indicate the cards(s) they are waiting on and why (for example, waiting on a needed level improvement, or wanting to take advantage of a new stack ability).
 - i. The general rule is that **players are allowed to build on stack improvements released in the same round**
 - ii. In the rare event that play becomes blocked by players waiting on each other:
 - 1. **first**, resolve in favor of technology requirements: in player order, players must decide whether they will release any eligible project(s) that increase technology levels; and
 - 2. **next**, resolve remaining projects: in player order, each player must decide whether to release any or all of their remaining projects.
 - iii. **Choosing not to release a project** that otherwise meets technology and GTM requirements **incurs a penalty of -1** 📉
- d. **Shelving projects** – when players are unable or unwilling to release a completed project they may “shelve” them, sliding the card one position along in the development space onto the printed bookshelf. Such projects are automatically eligible for release in future rounds, but they block further use of the development space. Only the printed dev requirement 🧑 on cards is “remembered” as having been completed; GTM, cost and technology requirements will need to be met to release them. Developers do not need to be assigned to shelved projects; however, players may still wish to assign them for card effects.

7. Next loop (if round 7, skip ahead to game end!)

- a) If it is round 3 or 5 that is ending, **resolve checkpoint effects** – in player order, trigger all card “at checkpoint” effects (**green powers**). *Full game only: also resolve checkpoint cards (see Full Game, below).*
- b) Return GTM counters to players and reset each player’s count of tokens to match their total count from printed ➡ icons on their development board, from the brand track and from their cards.
- c) Return any action markers or once-per-round markers on cards to players.
- d) Reset each player’s recruiter token to the (top) starting space on the recruit action.
- e) Advance the round marker.
- f) Advance the starting player token (pass to the left).

🚩 **Game end** – after round 7 ends, **Valuation**

- Final scoring: check and resolve ‘Game end’ card abilities, then add up all victory points:
 - **Stack VP** – the count of a player’s markers collected on all development boards
 - **Upgrade VP** – as shown by level markers on all upgradeable projects
 - **Card VP** – per VP icon (and explanatory card text), on all card types
 - **Venture Capital VP** – per VP icons on unflipped VC funding cards
 - **Track VP** – per VP shown on the brand and morale tracks
 - **Tucked cards** – 1VP per card tucked under an owned project (per card abilities)
 - **Cash flow VP** – subtract your total per-round staff costs from your total income: if positive, 1VP per \$5
- Resolve ties in favor of the owner of the highest leveled stack, then the player with the highest stack VP, then share the tie.



New Player Tips!

Notice that your company starts with **no income** – your starter project is a good candidate for an initial commercial offering, as it will provide some initial card income. Be cautious about rapidly hiring developers unless you can use them to develop products that will start generating income.

It's OK to let staff go to reduce costs, but take care to shore up team morale ☹️. In the full game, expect to lose 1-2 ☹️ each checkpoint as the strain of support costs catches up with you.

Releasing projects takes some planning – pay attention to **GTM** ➡️ as you start with just one, which limits what you can release to market until you earn or hire more.

Pay close attention to **technology level requirements** – some cards cannot be started until later! Be prepared for the \$ **costs** required to release projects, in addition to your recurring staff costs.

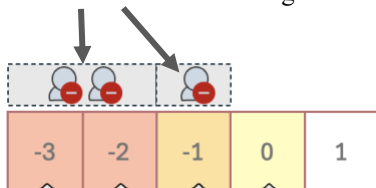
Full Game

Differences between the basic and full game are noted in the rules above. Here's a summary of the differences:

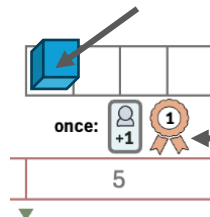
- As noted in **Setup**, players do not receive starter cards in the full game. Instead, players select only from randomly dealt cards for their starting hands. Players also start with \$10 rather than \$20.
- Players start one position further back on both morale and brand tracks (as marked by ☆).
- The full game allows **bidding** during the card market step under Planning
- Players may take **multiple recruit actions** during each round in the full game, but with rising demands on signing bonuses, as shown by the additional recruiting spaces on the round tracker. After the first action, players may only add to their developer hand if they first recruit and pay for at least one junior developer token.
- Once developer tokens are assigned to in-development project cards they may not be moved (except through card abilities such as Rosie's).
- In the full game, if a player *chooses* to shelve (pause) a project in development (see “Shelving projects” under Release) there is a morale penalty. (This penalty is avoided if a project is blocked from release due to missing technology requirements or go-to-market resources.)
- The full game's morale track includes additional bonuses and penalties (see below).
- **Checkpoints:** the full game includes **checkpoint cards** with additional checkpoint steps:
 - **Analyst review** – first, resolve the visible bonus awards reflecting industry and analyst enthusiasm. (*Solo game: resolve only the lower part of this card, which grants a money bonus according to the number of tags you have for the category shown.*)
 - **Cost-of-success** – then, reverse the card and resolve setbacks due to maintenance and compliance friction. Money penalties can be taken as deductions from the next round's income. (*Solo game: resolve only the top part of this card, which describes morale setbacks.*)
 - **Checkpoint abilities** – in player order, trigger all card “at checkpoint” effects (**green powers**), as in the basic game.

Guide to morale ☹️ track effects (penalties and bonuses) in the full game:

Attrition indicators – when morale is low, staff members may leave. See ‘Retention’ under Staffing.

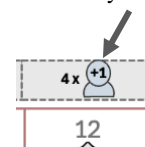


Once-per-game bonuses – when your tracker first reaches space 5 or 7, gain the bonus shown and mark one of the four spaces



Claim 1 developer card from anywhere in the talent row to your dev hand, plus 1 brand

High-morale developers become more productive: assign each +1 bonus to a (different) developer token of your choice every Release



After reaching the final space, gain \$3 per any additional morale received

Solo Game

The changes for a 1P game are noted in the rules above and on game boards / references; here's a summary of the differences plus some guidance on resolving card effects. While you can play either the basic or full game, the full game is recommended for a more challenging game.

- **Setup** (*full game*): Draw 8 tech cards and keep 4; draw 4 developers and keep 2 in your developer hand. Randomly select two different base projects (OS, DB, etc.) and mark them with a neutral player marker to create two initial neutral stacks. Start your **brand tracker at 1** instead of 0, as indicated by the ☆ on the brand track.
- From round 2 on, generate two **open source** updates at the beginning of each round, as described in 'Open source' on page 5.
- During the 'Card market' phase, **remove one card from an available market space** (to discard pile) before selecting a free card for the current turn. Do not mark the first card with a coin, and do not swap cards in the market after the end of your Actions during a turn.
- Similarly, once per round you may **discard one developer card** from one of the spaces available to you in the talent row. Do not mark the first developer card with a coin.
- In general, you may use card effects which require other players, by substituting the neutral player. You may assume that the neutral player has at least one of all tag types. If you need to reveal cards from the neutral player's hand, reveal them from the deck instead. So, you can take the rewards on Real-time Communication and Photo Followers, compare stacks using Civilization Strategy, and play the battle effect in Multiplayer Text Wars (revealing a card from the deck instead). If you play Swipe to Match on an  stack owned by the neutral player, you may gain the card's reward (because doing so meets the card's criteria).
- As noted under Full Game above, do not resolve the rewards/penalties marked as 2P+ during the checkpoint phase (skip the multi-player rewards and penalties).

Reference

About Card Effects and Abilities

The use of printed card effects or abilities is always optional except in cases where they may result in a penalty (e.g. Online Casino).

Card effects supersede game rules (for example, card abilities may allow you to release a project even when certain requirements are not met).

Card tags and printed bonuses (such as an increase in brand) are immediately activated or received when a card is played. If a card grants bonuses based on tags, that bonus will always credit any tags on the card itself.

If card powers cause multiple abilities to trigger at once, the player chooses the order in which they activate (for example, a card with a category tag might have a ‘when released’ ability while another card has effects that trigger when that tag is played – the player would choose the order in which those effects were resolved.)

Any card effect which modifies the deck or discard pile should **leave card order unchanged** except as directed in the text. Neither deck is free to be searched – only the top card of the discard pile is required to remain visible. When discarding multiple cards, players may decide the order in which they are placed on the discard pile.

If a player is required by a card effect to pay a money cost but has insufficient cash, they must flip the next available VC card and pay the cost. If they do not have a remaining VC card available, they do not have to pay penalties in card effects (such as losing money at the casino) but cannot choose to play cards or take actions that require spending money.

Card ability terminology (see also Terminology on the front page)

- **“Players on this stack”** – The set of players owning or having used a given stack, determined by possessing a marker for that stack in the matching technology area at the top of their player board (that is, having previously used the stack in question to satisfy requirements for a project card, or as part of a card effect)
- **“ card”** (**** etc.) – any project card that has a **requirement for this technology** (*does not need to be a stack/striped card*)
- **“ card”** (**** etc.) – any card (project or industry) marked with the indicated software category tag
- **** grants the ability to release projects at 1 under the printed value of any one technology requirement (for ****, **** etc.)
- **“small / medium / large projects”** – projects of length **** 1, 2 or 3 respectively
- **“a project”** – in card abilities this generally refers to a project you own, i.e. a project card you have previously released
- **“When released:”** or **“after releasing...”** – ability triggered when a project is successfully released to market (‘releasing’ always refers to projects)
- **“When played:”** – ability triggered at the time an industry card is played (usually during the Actions phase)
- **“At promotion:”** – ability triggered when a developer is promoted (developer card comes into play)
- **“At Checkpoint:”** – ability triggered during the checkpoints after rounds 3 and 5
- **“New action:”** – card ability which may be used by any player on the same stack. Using the ability consumes an action during the Action phase, and the player places a marker in one of the available slots when they use the action. Players may not use the same action card twice in a single round. The first space is reserved for (and must be used by) the card owner, and the third space is only available in a 4-player game.
- **“When releasing on this stack...”** – ongoing stack abilities (yellow text) add capabilities to a stack; these abilities will benefit future releases that use this stack but they do not benefit the card that itself added that ability.
- **“your hand”** – in general this refers to your hand of tech cards – **exclude standard projects and developer cards**

- “Discard” – discard a card from your main hand (to the discard pile) in return for a bonus. You can’t discard standard projects or developer cards for such effects. (On some conditions such as attrition, the rules can instruct you to discard a developer card to the developer discard pile.)
- “Draw a card” – draw a card from the **top of the deck**. If the deck is ever empty when a card needs to be drawn or revealed, first shuffle the discard pile into a new deck.
- “Reveal a card” – some cards instruct players to reveal a card to determine effects; draw from the top of the deck such that the card is visible to all players, evaluate the effect, then place the card on the discard pile (this does not count as discarding a card from hand)
- “Reveal cards until...” – reveal cards from the top of the deck, placing them in a new pile until a card of the indicated type is revealed (most commonly, to claim that card to the player’s hand.) In the less common event that the deck is exhausted, flip the discard pile over and continue revealing cards; if the discard pile is also exhausted, then the card ability has no further effect. After a search extends into the discard pile in this way, shuffle all revealed cards with any remaining cards in the discard pile to create a new deck (the discard pile will now be empty.)
- “Claim 1 card from the market” – take any card from the card market into your hand (ignoring the normal limits on which cards you can select during Planning), then refill the market. If the card has a coin on it, claim the coin as well.
- “Sell a card” – discard one card in return for a benefit (counts as discarding for any game effects related to discarding from hand)
- “After  [...]” – an effect or ability that you can activate after playing a card with the indicated tag. For projects, this happens when you first release the project to market. For developers, this happens when a card with a tag is promoted.
- “Tuck a card” – place a card underneath another project card as instructed by the text of the effect. Each card tucked in this way will score as 1VP at the end of the game. (The card you are tucking under cannot be a +1 or +2 improvement card as those are already partially tucked under base cards and do not have any owner markers.)

Standard Project clarifications

Prototype – Draw cards from the top of the deck.

Upgrade – There are various types of upgradeable cards in the game, including all starter technologies as well as projects like Cloud Computing or Instant Messaging (marked by ). By default, the Upgrade project allows you to advance level markers on just one or two of these cards per release; some cards (Linus and Feature Flags) allow you to target additional cards.

As explained in card text, the costs listed must be met **per card**. For example (using the full game version of the Upgrade card), you can advance Operating System to level 4 and Instant Messaging to level 2 with a single Upgrade if you have at least 3 dev strength  assigned *and* expend one GTM  and pay \$3 (because the 4th OS level space shows the  symbol, increasing the cost).

Players may upgrade their own cards or any card in the stack area, whether they are currently using a given stack or not. In this way, players can upgrade and then use a new stack in the same round.

Standard projects are subject to the same potential morale penalty when a player chooses not to release them, but there is not actually any need to leave a standard project on the player board when they are shelved (see “Shelving projects” under Release) as developers would need to be reassigned to supply any dev strength  anyway. Simply return standard projects to player areas after resolving any  penalty.

If two players attempt to upgrade the same card, and this leaves no available card for a second player to upgrade, the second player’s Upgrade returns to their player area without effect or penalty.

Project card clarifications

3D Visual FX – The cost discount on an industry card applies only as the card is initially played – a staff card’s recurring cost will need to be paid normally in subsequent rounds. This card’s stack ability is not added to the stack until the card is released, therefore it does not benefit from its own discount (in this case, on  tags).

Auction Site – For example, if you have 2  tags and reveal a 3 from the deck, you have the opportunity to sell 1 card for \$5. If you have 4  tags and reveal a 6, you can sell a card for \$8 (the maximum).

AI-Powered Learning Engine – This card’s stack ability is not added to the stack until the card is released, so it does not benefit from its own bonus on  tags. When using this ability with future releases, if the discard pile is searched and no  is found, the player may still choose the ‘draw 1 card’ option as an alternate reward.

Code Development Suite – This card scores VP according to  markers on the player board of this card’s *owner* (1VP for every 2), regardless of whether the owner also owns this stack or not.

Cloud Computing – This card’s stack ability is not added to the stack until the card is released, so it does not benefit from its own money bonuses.

Cloud Storage – This card’s ability activates any time a player *collects a marker* for this card’s stack for the first time. A consequence is that the reward is not received when this card is first played as a New Stack, since a marker for the new stack is not collected. In the case that the owner adds the card to their own existing  stack, players (including the owner) may already have a marker for that stack, and so won’t be eligible for the reward. (In the less common case that the owner plays into their own stack but hasn’t yet used that stack, the reward would trigger immediately since they are taking a marker by playing the card). For the game end effect, “owner of the original stack this card uses” refers to the color of the marker in the lower left used as the base for this stack (see ‘New Stack cards’ under Release.)

Deep Learning – This card’s stack ability is not added to the stack until the card is released, therefore it doesn’t benefit from its own discount (in this case, on  tags). For the VP award, the card owner scores the VP printed on the card (here, 2VP), and then *if it is a 3 or 4-player game* award an additional 2VP to a player on this stack per the card’s instructions. If there is a tie between two players in the VP winning condition, each receives 1VP. If there is a tie among more than 2 players, no VP are awarded.

Document Coauthoring – see Deep Learning for clarification of game end VP condition

Electronic Payments – see Deep Learning for clarification

Enterprise Data Manager – (Reference: Ellison’s involvement was revealed by court filings in a Fair Fight lawsuit against pro-Trump group True the Vote, and confirmed by group leader Catherine Engelbrecht and a participant)

Game Storefront – This card’s endgame bonus counts any non-stack (non-striped) projects you own where  is 1 (“small”).

Generative AI Coding – The project you advance must have appropriate  dev strength assigned. If this effect causes that project to meet the Release line, the project becomes eligible for release during the current Release phase. This card’s stack ability (to bypass tech requirements) becomes immediately available to all players on this stack once the card is released (naturally, this ability is free for the card owner to use.)

High-Frequency Trading – This card’s ability applies any time a game effect causes you to draw cards, including cards like Prototype, Travel Booking, Kaminsky and Sheridan (among others). If the game effect requires you to pay for each card you draw, you are not required to pay for the additional card you draw. Discard an additional card after choosing the card(s) you wish to keep (so that the total number of cards kept is unaffected.)

Image Recognition – A card must be discarded from a player’s hand to activate this card’s effect – revealing and discarding cards to test effects, or deciding not to keep some cards (e.g. after Prototype) does not trigger this card’s effect. Selling cards (discarding a card from hand for money) does activate this card’s effect.

Large Language Model – This card’s stack ability (to repeat a “when released” effect) becomes immediately available to players on this stack once the card is released, so can be used during the same Release phase (including by the owner, who could choose to repeat this card’s “when released” effect or that of another.) Players may choose a “when released” effect from any card they own.

Microblogging – (Reference: Rankings of disinformation discoverability from TrustLab, an independent AI-driven trust & safety company based in Palo Alto – specifically a 2023 study ranking the prevalence and sources of disinformation across social media platforms in several European countries, in support of the European Commission’s Code of Practice on Disinformation.)

Minimum Viable Product – The wild tag becomes available once the project is successfully released, and therefore cannot be used to trigger developer bonuses while it is in the process of being developed or released (abilities which trigger “after” do apply – and will always match tags because the card’s ability has become active)

Mobile Framework – see Deep Learning for clarification

Multimedia Player – Because the additional small project you can release has no developers directly assigned to it, you cannot use this card’s ability to release a standard project like Upgrade (no dev strength  → no effect.)

Future projects of  size 1 or 2 (small or medium) which use this new stack to fulfill  requirements receive the printed discount on their release cost. Note that this card must use a different stack to fulfill *its*  requirement (the stack that provides the base level of this new stack,) so this card is not eligible for its own discount. (The Multimedia Player engine becomes an easier way to build and distribute *new* projects – see ‘New Stack cards’ under Release.)

Multitasking – see Multimedia Player for clarification on the immediate release of a small project.

Multiplayer Text Wars – Players may not show the revealed card to other players, only to each other. Remember not to include standard projects or developer cards in your player hand. Players use the cost from the card they own (selected at random by their opponent) in the battle (and keep the card).

Remote Desktop – ‘card action’ refers to “New action” powers on other stack cards; this card’s action allows you to use or repeat another card’s action even if you have already used it, or when other players have already marked all of the available action spaces.

Replatform – This card lets you replace markers collected in the technology-specific pools at the top of your development board, for example to replace markers that benefit other players with your own markers (increasing stack income and VP), or to gain access to card abilities offered by a different stack. The replacement markers must be for the same technology type. Newly played or newly improved stacks may be used for the replacement markers, per the normal rules (see “meet technology requirements” under Release.) See similar clarifications regarding per card costs (here, per marker) under ‘Upgrade’ above.

Social APIs – If players agree, feel free to allow the ‘Scroll’ action to activate the “when played” effect on non-project cards (i.e., Reese) as well.

Social Network Space – see Social APIs. (Reference: Proceedings of the 21st ACM Internet Measurement Conference, Edelson et al, “Understanding Engagement with U.S. (Mis)Information NewsSources on Facebook”)

Subscription Analytics – If your current brand is at the minimum position on the track (-2), you have no brand to trade via the ‘Freemium’ action.

Tech Leadership Program – The free promotion is in addition to any other promotion(s) in the current round.

To-Do Organizer – These ongoing abilities become active for any player on the stack as soon as they have the required number of  shown, per ability. With the second ability, players can take 3 actions in a row during the Actions phase of each round, and can release 3 projects in a row during the Release phase. For a 5th development space, use the space beneath your player board.

Virtual 3D Engine – see Deep Learning for clarification on game end VP condition

Virtual Assistant – All players gain the card’s ability when playing  on this stack (including the owner when the card is played). There is no limit on how many times a type of task may be ‘purchased’. For example, when a player gains their fourth  on a project card, and use this stack, they might choose to claim the top two cards of the discard pile. Note that the third option can activate checkpoint effects on non-project cards such as Fei-Fei, or on a stack card like Large Language Model (if it is on a stack that the player is also on, in which case the player does not need to own that card to use its ability via Virtual Assistant).

Developer card clarifications

Important: blue text abilities on developer cards are only active when a developer is *assigned* to a project (and when that project matches any required criteria). For example, Liskov grants a \$4 discount “when releasing

DB”: this means that if a project shows a **DB requirement** and is released with (promoted/Senior) Liskov assigned, the \$ cost shown on the project is reduced by \$4 (not below \$0).

Some developers have ongoing yellow text abilities that are not related to their assigned projects and always available (Hal, Lotfi, Parisa, Radia, Lynn, Susy Thunder).

Charlie – When you use Charlie’s ‘at promotion’ ability, his project is advanced to the release line and becomes eligible for release in the current round.

Hal – See notes on the project High-Frequency Trading. Once promoted, Hal’s ability is always active.

Gaben – Gaben’s ability is triggered when releasing *projects* with the indicated tags, so it isn’t activated by Gaben’s own .

János – If both projects that are releasing have 'when released' abilities, you choose the order in which they activate.

Linus – When assigned to an Upgrade or Replatform project, Linus gains +1 dev strength  and you may target one additional card or marker using the Upgrade or Replatform (by default, increasing the number of targeted updates / cube changes from two to three; see Upgrade and Replatform)

Industry card clarifications

Adele and Edsger – The ability to reduce  dev requirements on projects in development extends through release – for example, a single developer on an Upgrade project would be able to make two eligible upgrades.

Alessandro – The ‘once per round’ ability may be used during any part of a round (including Actions and Release). The marker is a reminder (not an action); clear it at the end of every round.

Casey – See Alessandro for clarification of the ‘once per round’ ability. Because playing Casey gains a morale, their ability can be immediately triggered to gain an additional morale for the round in which the card is played.

Jing – with Jing in play, you release all  projects at -1 GTM  cost. For example, Junk Mail Filter normally requires GTM of 1 to release, but with Casey the project would be released without GTM cost. There is no limit to the number of projects that can receive this discount in a given round.

Kevin – see Jing

Laura – see Jing

Marian – this card’s money bonus only triggers when increasing the level on an “existing” stack, so is not triggered when playing a base or new platform card to create a new stack. It is however triggered when a new platform card is played as a stack card on an owner’s existing stack of the same type, since in that case the new platform card just increases the existing stack’s level.

PJ – See notes on the project High-Frequency Trading

Product Placement – The card grants +1 GTM  when played. After you play your third , gain an additional GTM counter, reflecting the increase in content now available to you for product placement. If you later drop below 3  (for example, due to attrition), you also lose the additional counter.

Reese – see Jing

Rhea – see Alessandro for clarification of the ‘once per round’ ability.

Shelby – see Jing

Sheridan – Count your brand level above 0 (for example, if your brand is 5, you would draw 2 cards); always draw at least one card

Viral Advertisements – See Product Placement

Checkpoint clarifications

Stack penalties and bonuses are assigned per player; players do not incur repeated bonuses or penalties for being on multiple qualifying stacks (or being on qualifying stacks multiple times).

FAQ

Is there a Planning phase in the first round? Yes, the first round begins with Planning, and players select an additional card (limited by their brand to the first two spaces in the market).

Can you assign more developers to a project than required or useful? Yes, but once assigned they cannot be reassigned later in the same round (per normal rules.)

What happens if a player moves or loses a developer card that had been assigned to a project which is not yet complete? If the player cannot assign sufficient  dev strength, the project is immediately returned to hand (any development progress is lost).

What happens if a player's available GTM  changes mid-release (for example due to positive or negative changes in brand)? Add GTM counters as they are gained and remove them from the player's pool of available counters when they are lost. If, for example, the last project released in a round causes a reduction in GTM and the player has no counters to lose, the reduction will only affect future rounds (because players sync up the number of GTM counters they have with their available GTM at the end of each round).

What happens if a player plays a New Stack card on their own stack? The card is treated as an improvement to that existing stack and can be added to it, per "Place the played project into the appropriate game area" under Release.

When the base stack that a New Stack card is built on increases its level, does the new stack also increase in level? Yes, because the level of the new stack card is always determined by adding the level of the base stack to the level shown on the new stack card.

Is it necessary to collect a stack marker when using a neutral player stack (in a 2P or 3P game)? Yes, for purposes of evaluating game effects, including determining which players have access to stack abilities or actions and use of cards like Replatform.